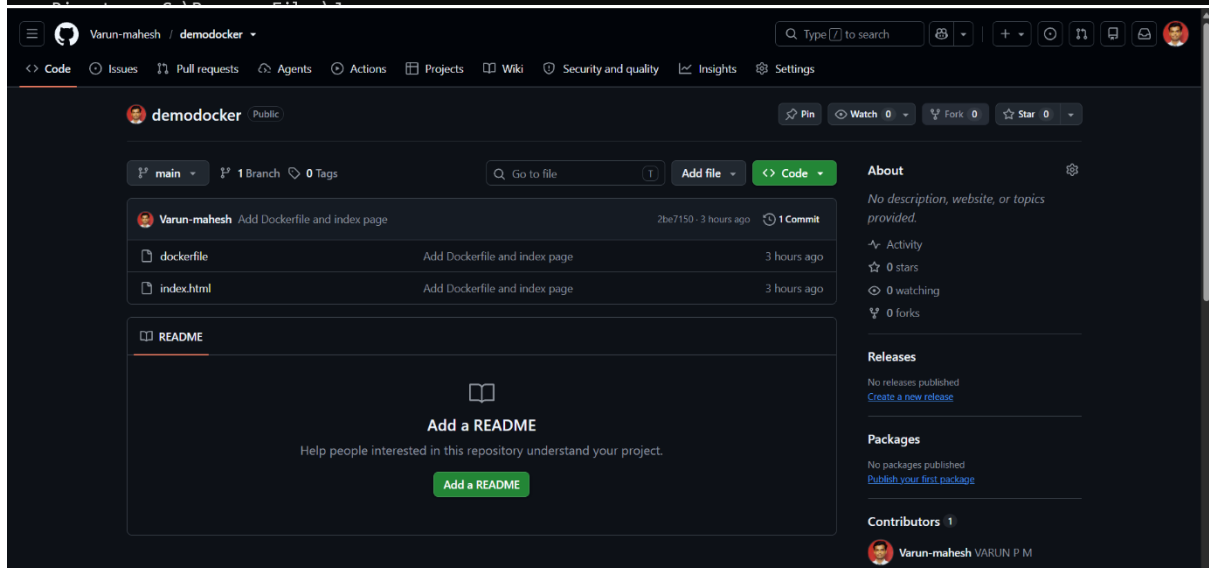


```
services:
  frontend:
    image: httpd:2.4
    container_name: frontend
    ports:
      - "8080:80"
    volumes:
      - ./frontend:/usr/local/apache2/htdocs/
    networks:
      - app-network
  backend:
    image: php:8.2-apache
    container_name: backend
    ports:
      - "8081:80"
    networks:
      - app-network
  database:
    image: mysql:8.0
    container_name: database
    environment:
      MYSQL_ROOT_PASSWORD: root
      MYSQL_DATABASE: myapp
      MYSQL_USER: appuser
      MYSQL_PASSWORD: apppassword
    volumes:
      - mysql-data:/var/lib/mysql
    networks:
      - app-network
networks:
  app-network:

volumes:
  mysql-data:
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> & "C:\Users\Varun Mahesh\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\25\bin\java.exe" -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)
VARUN-MAHESH C:\Users\Varun Mahesh>
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> java -version
java version "26" 2026-03-17
Java(TM) SE Runtime Environment (build 26+35-2893)
Java HotSpot(TM) 64-Bit Server VM (build 26+35-2893, mixed mode, sharing)
VARUN-MAHESH C:\Users\Varun Mahesh> where.exe java
C:\Program Files\Common Files\Oracle\Java\javapath\java.exe
C:\Program Files (x86)\Common Files\Oracle\Java\java8path\java.exe
C:\Users\Varun Mahesh\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\25\bin\java.exe
VARUN-MAHESH C:\Users\Varun Mahesh> Get-ChildItem "C:\Program Files\Java" -ErrorAction SilentlyContinue
```



```

MINGW64 C:\Users\Varun Mahesh\Desktop\task
dockerfile index.html task-backup.tar

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git init
Initialized empty Git repository in C:/Users/Varun Mahesh/Desktop/task/.git/

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git add dockerfile index.html
warning: in the working copy of 'index.html', LF will be replaced by CRLF the next time Git touches it

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   dockerfile
        new file:   index.html

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        task-backup.tar

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git commit -m "Add Dockerfile and index page"
[master (root-commit) 2be7150] Add Dockerfile and index page
 2 files changed, 14 insertions(+)
 create mode 100644 dockerfile
 create mode 100644 index.html

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git remote add origin https://github.com/Varun-mahesh/demodocker.git

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git remote -v
origin https://github.com/Varun-mahesh/demodocker.git (fetch)
origin https://github.com/Varun-mahesh/demodocker.git (push)

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (master)
$ git branch -M main

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (main)
$ git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (4/4), 529 bytes | 529.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
to https://github.com/Varun-mahesh/demodocker.git
 * [new branch] main -> main
branch 'main' set up to track 'origin/main'.

varun Mahesh@VARUN-DHARUN MINGW64 ~/Desktop/task (main)
$

```

```

},
  "Image": "sha256:ef3388122a742b558989b8f46a9a9ca6fe4220ea90d7ef63e3b5b5cb390c2265",
  "ResolveConfPath": "/var/lib/docker/containers/9fb9fd0e1b87c92fcabc03b973db09e0b42d27f94eae8b3dd963cb227f8adc05/resolve.conf",
  "HostnamePath": "/var/lib/docker/containers/9fb9fd0e1b87c92fcabc03b973db09e0b42d27f94eae8b3dd963cb227f8adc05/hostname",
  "HostsPath": "/var/lib/docker/containers/9fb9fd0e1b87c92fcabc03b973db09e0b42d27f94eae8b3dd963cb227f8adc05/hosts",
  "LogPath": "/var/lib/docker/containers/9fb9fd0e1b87c92fcabc03b973db09e0b42d27f94eae8b3dd963cb227f8adc05/9fb9fd0e1b87c92fcabc03b973db09b42d27f94eae8b3dd963cb227f8adc05-json.log",
  "Name": "/task-web",
  "RestartCount": 0,
  "Driver": "overlayfs",
  "Platform": "linux",
  "MountLabel": "",
  "ProcessLabel": "",
  "AppArmorProfile": "",
  "ExecIDs": null,
  "HostConfig": {
    "Binds": null,
    "ContainerIDFile": "",
    "LogConfig": {
      "Type": "json-file",
      "Config": {}
    }
  },

```

VARUN-MAHESH C:\Users\Varun Mahesh> docker inspect task-web

```

[
  {
    "Id": "9fb9fd0e1b87c92fcabc03b973db09e0b42d27f94eae8b3dd963cb227f8adc05",
    "Created": "2026-09-29T03:24:13.210246349Z",
    "Path": "httpd-foreground",
    "Args": [],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 486,
      "ExitCode": 0,
      "Error": "",
      "StartedAt": "2026-09-29T03:24:13.557139128Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    }
  }
]

```

```
Total reclaimed space: 94.21kB
VARUN-MAHESH C:\Users\Varun Mahesh> docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
VARUN-MAHESH C:\Users\Varun Mahesh> docker images
```

| IMAGE              | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|--------------------|--------------|------------|--------------|-------|
| httpd:2.4          | 03f858efb82c | 177MB      | 47.6MB       |       |
| httpd:latest       | 03f858efb82c | 177MB      | 47.6MB       |       |
| node:latest        | a723b54c35a7 | 1.8GB      | 461MB        |       |
| task-backup:latest | 560a8dc2c7f8 | 175MB      | 45.3MB       |       |
| task-web:latest    | ef3388122a74 | 175MB      | 45.3MB       |       |

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network ls
```

| NETWORK ID   | NAME   | DRIVER | SCOPE |
|--------------|--------|--------|-------|
| 2664082c834a | bridge | bridge | local |
| e506cd2527af | host   | host   | local |
| d0298fdc2ead | none   | null   | local |

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network create sjc-demonet
Fa0965a3079d042eb84362e3b3316320678027f3fee5904d7abe63335c608632b
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network ls
```

| NETWORK ID   | NAME   | DRIVER | SCOPE |
|--------------|--------|--------|-------|
| 2664082c834a | bridge | bridge | local |

```
RAM 0.98 GB   CPU 0.08%   Disk: 3.33 GB used (limit 1006.85 GB)
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker system prune
```

WARNING! This will remove:

- all stopped containers
- all networks not used by at least one container
- all dangling images
- unused build cache

Are you sure you want to continue? [y/N] y

Deleted Containers:

```
3242b5f23f1f2833afd4cf9ae2ed2fe958e0b4721a18b3e8f3492abcb78443cb
d08eb118e879cfa82b8ba4ba30fb787b820cb2e5a32286ce8e677171547ff830
```

Deleted Networks:

```
sjc-demonet
```

Deleted build cache objects:

```
j5mh2phg2pygrfeabo66jw79t
aizph6g9reag3lwlgn4h6n1a
nqq69h63966y0aahsi1ay7c7g
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network ls
```

| NETWORK ID   | NAME   | DRIVER | SCOPE |
|--------------|--------|--------|-------|
| 2664082c834a | bridge | bridge | local |
| e506cd2527af | host   | host   | local |
| d0298fdc2ead | none   | null   | local |

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network create sjc-demonet
```

```
24e1a633cc7486540901e27ed6c2c34f373ad2c22925deea4cf5d7faec87c143
```

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker network ls
```

| NETWORK ID   | NAME        | DRIVER | SCOPE |
|--------------|-------------|--------|-------|
| 2664082c834a | bridge      | bridge | local |
| e506cd2527af | host        | host   | local |
| d0298fdc2ead | none        | null   | local |
| 24e1a633cc74 | sjc-demonet | bridge | local |

```
VARUN-MAHESH C:\Users\Varun Mahesh> docker system prune
```

WARNING! This will remove:

```
VARUN-MAHESH C:\Users\Varun Mahesh> & "C:\Users\Varun Mahesh\AppData\Roaming\Code\User\globalStorage\pleiades.java-extension-pack-jdk\java\25\bin\java.exe" -version
openjdk version "25.0.4.1" 2026-08-18 LTS
OpenJDK Runtime Environment Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS)
OpenJDK 64-Bit Server VM Temurin-25.0.4.1+1 (build 25.0.4.1+1-LTS, mixed mode, sharing)
VARUN-MAHESH C:\Users\Varun Mahesh> |
```

#### Getting Started

## Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

`C:\ProgramData\Jenkins\jenkins\secrets\initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

Continue

Day - 4

Docker network:

Docker network is

docker network create sjc-demonet (we can create multiple bridge network)

docker run -

docker system prune -a

(pruning cut off the system → remove the network idel).

docker run -d --name demo -p 8081:80 --network-sjc-demonet customing

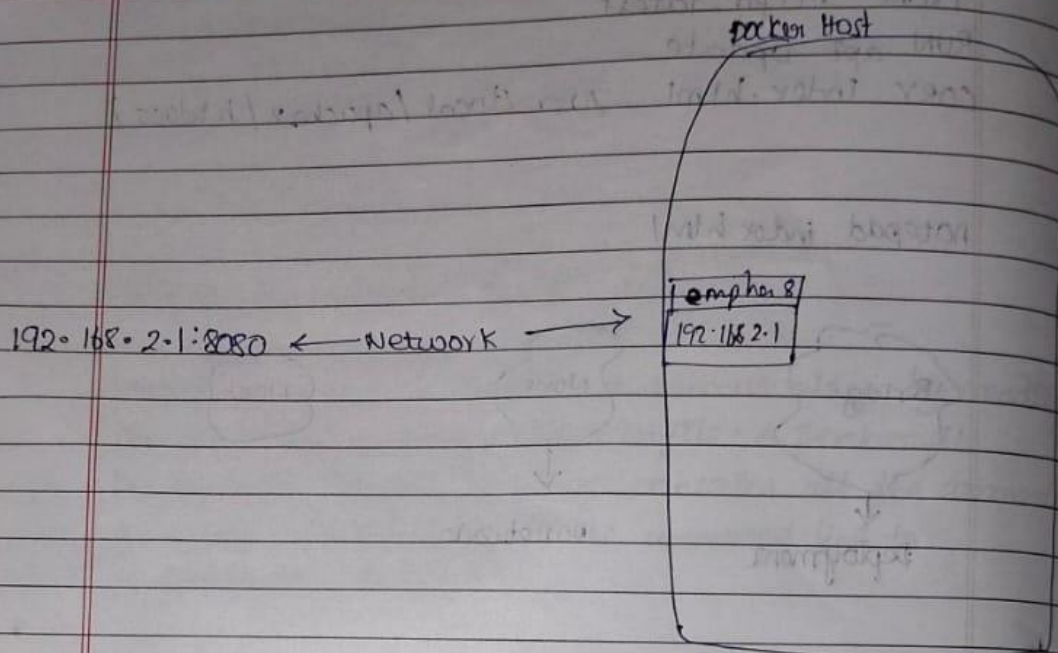
docker inspect demo

docker network inspect sjc-demonet

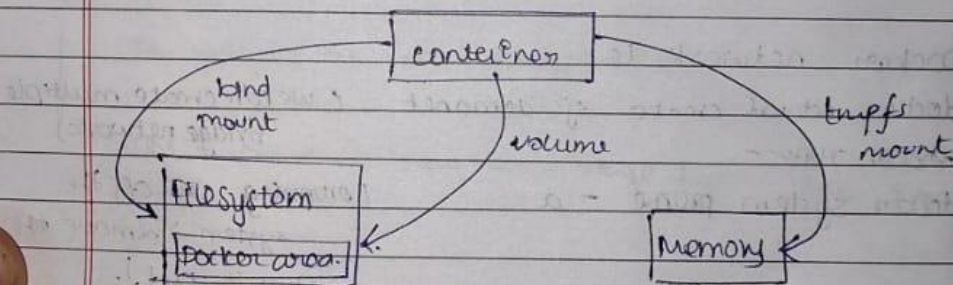


Date .....

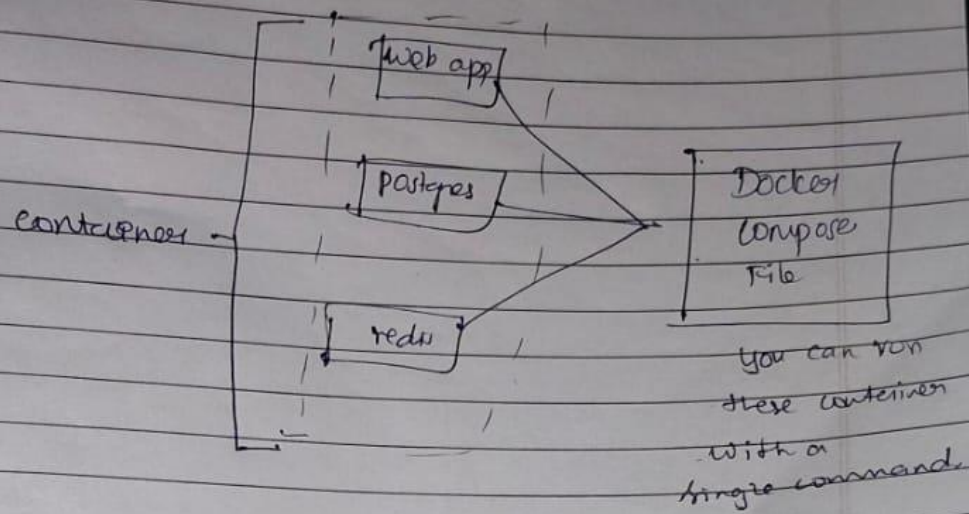
Page .....



Types of Docker volume :



Docker compose makes it easier to configure and run application made up of multiple containers. For example : imagine being able to define three containers - one running a web app, another running postgres and a third running redis - all in one YAML file and then running those three connected containers with a single command.



notepad docker-compose

rename docker-compose.txt docker-compose.yaml

Purpose of Docker compose:

→ to run multiple docker containers together using one configuration file.

notepad index.html

mv index.html.txt index.html

explorer.

docker compose up

notepad docker-compose.yaml

docker compose images

docker compose network

docker compose down

docker compose up -d